

```
!pip install transformers sentencepiece langdetect nltk sentence-transformers scikit-learn
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dist-packages (0.2.1)
Collecting langdetect
  Downloading langdetect-1.0.9.tar.gz (981 kB)
    981.5/981.5 kB 9.2 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: sentence-transformers in /usr/local/lib/python3.12/dist-packages (5.4.0)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.10.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
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Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.0)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages (from langdetect) (1.17.0)
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Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (2.10.0+cpu)
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Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: fspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (2023.5.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (1.2.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.27.0)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.12.1)
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Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (3.1.4)
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Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (2025.11.11)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (1.0.7)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (3.10)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (0.14.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.11.0->sentence-transformers) (1.3.0)
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Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (0.0.2)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2) (3.0.2)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers) (2.19.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->transformers) (0.1.2)
Building wheels for collected packages: langdetect
  Building wheel for langdetect (setup.py) ... done
  Created wheel for langdetect: filename=langdetect-1.0.9-py3-none-any.whl size=993223 sha256=701d63ccbc2cda29169e6b07252ea0c23b
  Stored in directory: /root/.cache/pip/wheels/c1/67/88/e844b5b022812e15a52e4eaa38a1e709e99f06f6639d7e3ba7
Successfully built langdetect
Installing collected packages: langdetect
Successfully installed langdetect-1.0.9
```

```
# Core libraries
from transformers import M2M100ForConditionalGeneration, M2M100Tokenizer
from langdetect import detect
from nltk.translate.bleu_score import sentence_bleu
from sentence_transformers import SentenceTransformer
from sklearn.metrics.pairwise import cosine_similarity
import numpy as np
import time
```

```
# Load translation model (supports 100+ languages)
model_name = "facebook/m2m100_418M"

tokenizer = M2M100Tokenizer.from_pretrained(model_name)
model = M2M100ForConditionalGeneration.from_pretrained(model_name)

# Load embedding model for semantic similarity
embedder = SentenceTransformer('all-MiniLM-L6-v2')

print("Model loaded successfully")
```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
tokenizer_config.json: 100% 298/298 [00:00<00:00, 9.44kB/s]

vocab.json: 3.71M/? [00:00<00:00, 33.1MB/s]

sentencepiece.bpe.model: 100% 2.42M/2.42M [00:00<00:00, 12.2MB/s]

special_tokens_map.json: 1.14k/? [00:00<00:00, 68.4kB/s]

config.json: 100% 908/908 [00:00<00:00, 12.8kB/s]

pytorch_model.bin: 100% 1.94G/1.94G [00:38<00:00, 107MB/s]

model.safetensors: 100% 1.94G/1.94G [00:28<00:00, 169MB/s]

Loading weights: 100% 512/512 [00:04<00:00, 111.00it/s, Materializing param=model.shared.weight]

The tied weights mapping and config for this model specifies to tie model.shared.weight to lm_head.weight, but both are present
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, bu
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, bu
generation_config.json: 100% 233/233 [00:00<00:00, 3.78kB/s]

modules.json: 100% 349/349 [00:00<00:00, 3.82kB/s]

config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 1.30kB/s]

README.md: 10.5k/? [00:00<00:00, 85.4kB/s]

sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 574B/s]

config.json: 100% 612/612 [00:00<00:00, 8.25kB/s]

model.safetensors: 100% 90.9M/90.9M [00:03<00:00, 219MB/s]

Loading weights: 100% 103/103 [00:00<00:00, 189.99it/s, Materializing param=pooler.dense.weight]

BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key | Status | |
-----+-----+---
embeddings.position_ids | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

tokenizer_config.json: 100% 350/350 [00:00<00:00, 6.26kB/s]

vocab.txt: 232k/? [00:00<00:00, 3.47MB/s]

tokenizer.json: 466k/? [00:00<00:00, 2.21MB/s]

special_tokens_map.json: 100% 112/112 [00:00<00:00, 1.24kB/s]

config.json: 100% 190/190 [00:00<00:00, 1.68kB/s]

Model loaded successfully

```

```

def translate(text, src_lang="en", tgt_lang="fr"):
    """
    Translate a single sentence
    """
    tokenizer.src_lang = src_lang

    encoded = tokenizer(text, return_tensors="pt")

    generated_tokens = model.generate(
        **encoded,
        forced_bos_token_id=tokenizer.get_lang_id(tgt_lang)
    )

    return tokenizer.batch_decode(generated_tokens, skip_special_tokens=True)[0]

```

```

def translate_paragraph(text, src="en", tgt="fr"):
    """
    Translate paragraph by splitting into sentences
    """
    sentences = text.split(".")

```

```

translated = []

for sentence in sentences:
    if sentence.strip():
        translated.append(translate(sentence.strip(), src, tgt))

return ". ".join(translated)

```

```

def is_complex(text):
    """
    Detect whether text is complex using semantic similarity
    """
    sentences = text.split(".")

    if len(sentences) < 2:
        return False

    embeddings = embedder.encode(sentences)

    similarities = []
    for i in range(len(embeddings) - 1):
        sim = cosine_similarity([embeddings[i]], [embeddings[i+1]])[0][0]
        similarities.append(sim)

    avg_similarity = np.mean(similarities)

    # Lower similarity → multiple contexts → complex
    return avg_similarity < 0.5

```

```

def smart_translate(text, target_lang="fr"):
    """
    Automatically:
    - Detect language
    - Decide sentence vs paragraph
    """
    detected_lang = detect(text)
    print("Detected Language:", detected_lang)

    # Decision logic
    if is_complex(text) or len(text.split()) > 15 or "." in text:
        print("[DECISION] Mode: Paragraph Translation")
        print("[INFO] Using function: translate_paragraph()")
        result = translate_paragraph(text, detected_lang, target_lang)
    else:
        print("[DECISION] Mode: Sentence Translation")
        print("[INFO] Using function: translate()")
        result = translate(text, detected_lang, target_lang)

    return result

```

```

def evaluate_bleu(reference, translated):
    """
    BLEU score for lexical similarity
    """
    ref = [reference.split()]
    cand = translated.split()

    return sentence_bleu(ref, cand)

def evaluate_embedding(reference, translated):
    """
    Semantic similarity using embeddings
    """
    emb1 = embedder.encode(reference)
    emb2 = embedder.encode(translated)

    sim = cosine_similarity([emb1], [emb2])
    return sim[0][0]

```

```

def measure_speed(text, target_lang):
    """
    Measures translation time using smart_translate
    """

```

```

start = time.time()
result = smart_translate(text, target_lang)
end = time.time()

return result, round(end - start, 3)

```

```

valid_langs = ["en", "fr", "de", "hi", "es", "zh", "ar"]

while True:
    print("\n==== AI Translator Menu =====")
    print("1. Translate Text")
    print("2. Exit")

    choice = input("Enter choice: ")

    if choice == "2":
        print("Exiting system")
        break

    elif choice == "1":
        text = input("\nEnter text: ")
        target_lang = input("Enter target language (fr/de/hi/es/en): ")

        if target_lang not in valid_langs:
            print("Invalid language code")
            continue

        try:
            # Use reusable timing function
            translated, exec_time = measure_speed(text, target_lang)

            print("\nTranslated Output:", translated)
            print("Translation Time:", exec_time, "seconds")

            # Evaluation choice
            eval_choice = input("\nDo you want evaluation? (yes/no): ").lower()

            if eval_choice == "yes":
                ref = input("Enter reference translation: ")

                if ref.strip():
                    print("\nChoose evaluation method:")
                    print("1. BLEU Score")
                    print("2. Embedding Similarity")
                    print("3. Both")

                    eval_type = input("Enter choice: ")

                    if eval_type == "1":
                        bleu = evaluate_bleu(ref, translated)
                        print("BLEU Score:", round(bleu, 4))

                    elif eval_type == "2":
                        emb = evaluate_embedding(ref, translated)
                        print("Embedding Similarity:", round(emb, 4))

                    elif eval_type == "3":
                        bleu = evaluate_bleu(ref, translated)
                        emb = evaluate_embedding(ref, translated)

                        print("BLEU Score:", round(bleu, 4))
                        print("Embedding Similarity:", round(emb, 4))

                    else:
                        print("Invalid evaluation choice")

            except Exception as e:
                print("Error:", e)

```

```

==== AI Translator Menu =====
1. Translate Text
2. Exit
Enter choice: 1

```

```

Enter text: The rapid advancement of technology has transformed the way people communicate, work, and interact, creating both
Enter target language (fr/de/hi/es/en): de

```

```
Detected Language: en
[DECISION] Mode: Paragraph Translation
[INFO] Using function: translate_paragraph()

Translated Output: Der schnelle Fortschritt der Technologie hat die Art und Weise verändert, wie Menschen kommunizieren, arbeit
Translation Time: 15.548 seconds

Do you want evaluation? (yes/no): yes
Enter reference translation: Der rasante Fortschritt der Technologie hat die Art und Weise verändert, wie Menschen kommuniziere

Choose evaluation method:
1. BLEU Score
2. Embedding Similarity
3. Both
Enter choice: 3
BLEU Score: 0.6901
Embedding Similarity: 0.8535

===== AI Translator Menu =====
1. Translate Text
2. Exit
Enter choice: 1

Enter text: While machine learning models can identify patterns in large datasets, their performance heavily depends on data qu
Enter target language (fr/de/hi/es/en): hi
Detected Language: en
[DECISION] Mode: Paragraph Translation
[INFO] Using function: translate_paragraph()

Translated Output: जबकि मशीन सीखने के मॉडल बड़े डेटासेट में पैटर्न की पहचान कर सकते हैं, उनकी प्रदर्शन डेटा की गुणवत्ता और उचित पैरामीटर ट्यूनिंग
Translation Time: 17.499 seconds

Do you want evaluation? (yes/no): yes
Enter reference translation: हालांकि मशीन लर्निंग मॉडल बड़े डेटा सेट में पैटर्न पहचान सकते हैं, उनका प्रदर्शन काफी हद तक डेटा की गुणवत्ता और उचित

Choose evaluation method:
1. BLEU Score
2. Embedding Similarity
3. Both
Enter choice: 3
BLEU Score: 0.3361
Embedding Similarity: 0.9629

===== AI Translator Menu =====
1. Translate Text
2. Exit
Enter choice: 1

Enter text: The bank near the river, which was recently renovated to improve customer services, attracts both tourists and loci
```